



Machine Learning II

Data Science M.Sc.

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Missing Data



Introductory Example

Diabetes data set:

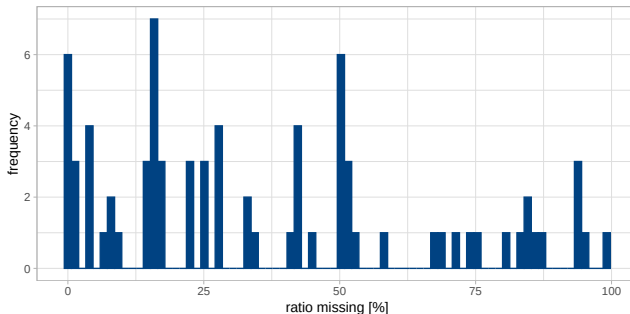
- ▶ In ML 1 you used a data set called *Diabetes* to develop a logistic regression classifier.
- ▶ The data set comes from the National Health and Nutrition Examination surveys (NHANES) in the USA. The full data contains 10000 elements and 76 features.



Introductory Example

Diabetes data set:

- ▶ In ML 1 you used a data set called *Diabetes* to develop a logistic regression classifier.
- ▶ The data set comes from the National Health and Nutrition Examination surveys (NHANES) in the USA. The full data contains 10000 elements and 76 features.
- ▶ **Problem:** there are a lot of missing values:





Introductory Example (cont)

Solution(?): we ignore all elements where we don't have complete observations

```
# the data set is named NHANES  
complete.cases(NHANES) |> sum() # read the help page for complete.cases
```

```
[1] 0
```

```
na.omit(NHANES) |> dim() # read the help page for na.omit
```

```
[1] 0 76
```

That's not what we want !!!



Imputation of missing values

Goal:

- ▶ Imputation is the name for any method which estimates missing data based on observed data to fill the gaps.
- ▶ The aim of imputing missing values is **not** to exactly predict the value that is missing.
- ▶ The aim is to complete the data so that the entire data can be used, therefore increasing the model accuracy by reducing the variance ($\sqrt{N}$ -law).
- ▶ The imputation process should not bias the estimates and predictions of a machine learning method.



Types of missing data

- ▶ **Missing completely at random (MCAR):** The probability that a missing value occurs¹ is constant. The occurrence is independent from any other variable in the data.
- ▶ **Missing at random (MAR):** The probability is not constant, but can be fully explained using variables in the data set.
- ▶ **Missing not at random (MNAR):** The probability that a missing value occurs, depends on non observed variables.

¹missing value occurs is often called *missingness*.



Types of missing data (cont)

Examples:

- ▶ **MCAR:** A hospital wants to record blood pressure (BP) for all its patients, but the measuring device happens not to work in 10% of cases. There is no systematic pattern to explain whether the device fails or not.
- ▶ **MAR:** A hospital records blood pressure (BP) for all its private patients, but only on 60% of patients with state insurance. The probability of missingness in BP is not constant for all patients but the probability of missingness is constant once the type of patient insurance is known.
- ▶ **MNAR:** In a particular clinical study, patients who were given a new treatment were more likely to get headaches and drop out of the study. If the side effect information is not recorded, then the data collected at the end of the study is missing not at random (MNAR).



Missing data and bias

- For MCAR or MAR data then omitting the data should not bias the results, but we will lose prediction power. The variance of our estimates will increase, and so will the mean squared error (MSE).



Missing data and bias

- ▶ For MCAR or MAR data then omitting the data should not bias the results, but we will lose prediction power. The variance of our estimates will increase, and so will the mean squared error (MSE).
- ▶ There is a special case of missing not at random (MNAR) which can be a particular problem. When the missing status of a variable depends on the value of that variable itself.
Example: Obese people are less likely to report their weight. The average weight derived from the data will be biased to lower values
- ▶ It is usually difficult to assess whether the missing data is MCAR, MCAR or MNAR.



Univariate Imputation Methods: Fixed methods

Mean replacement

- ▶ For each variable independently, calculate the mean of the non-missing values and set the missing values equal to that mean.
- ▶ This method is not good because the standard deviation of the imputed variable will decrease. The correlation with other variables will also decrease.



Mean replacement: example

NHANES data: pulse rate

```
summary(NHANES$Pulse)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
40.00	64.00	72.00	73.56	82.00	136.00	1437

```
mean(NHANES$Pulse, na.rm = TRUE)
```

```
[1] 73.55973
```

```
sd(NHANES$Pulse, na.rm = TRUE)
```

```
[1] 12.15542
```

```
NHANES$PulseImpMean <- NHANES$Pulse  
NHANES$PulseImpMean[is.na(NHANES$Pulse)] <- mean(NHANES$Pulse, na.rm = TRUE)
```

```
mean(NHANES$PulseImpMean, na.rm = FALSE)
```

```
[1] 73.55973
```

```
sd(NHANES$PulseImpMean, na.rm = FALSE)
```

```
[1] 11.24811
```



Univariate Imputation Methods: Fixed methods (cont)

Missing as a category

- ▶ For a **categorical variable** add a new level called "missing". E.g. for gender define three levels: `female`, `male` and `missing`. This is a reasonable a method.
- ▶ For a **numeric variable** replace the missing values with the mean or median value, and create a new indicator variable called `missing`. This is not a good method unless the number of missing values is small. It can lead to biased coefficients for the other variables. (You then could model an interactions with the indicator variable.)



Using logic rules for imputation

Sometimes a value can be imputed exactly or with high accuracy.

Examples:

- ▶ if a person's `weight` and `BMI` is known, but `height` is missing, then it can be calculated from the definition of body mass index.
- ▶ If a variable is number of `pregnancies` and a positive number is given then a missing value for `gender` can be imputed as female.
- ▶ Only 1% of breast cancer cases are in males. This means that if there is an indication of current or previous breast cancer, it is *acceptable* to impute that the `gender` is female.



Univariate imputation methods: Random methods

Mean/Variance Simulation:

- ▶ Suppose the variable x_1 has missing values.
- ▶ Compute the mean $\bar{x}_1$ and the standard deviation s_{x_1} of the known values.
- ▶ Replace the missing values with simulations from a normal $N(\bar{x}_1, s_{x_1}^2)$ distribution.
- ▶ This is a simple method and is effective provided x_1 is roughly normal distributed. If the distribution is very non-normal, then the variable after imputation will become closer to normally distributed.



Univariate imputation methods: Random methods (cont)

Direct Random Sampling:

- ▶ Sample the non missing values with replacement, to fill in the missing values.
- ▶ This approach is good when the missingness is independent of all other variables. Compared to the *mean/variance simulation* method, direct random sampling will preserve the distributional properties of the variable, and can be used for all data types.



Univariate vs multivariate methods

- ▶ The disadvantage with all univariate methods, is that they do not consider any co-dependencies with other variables.
- ▶ With most data there is codependency (correlation) between some variables, which can and should be incorporated into our imputation method.
- ▶ E.g. If we know a person's `height`, `age` and `gender`, then we can impute the `weight` much more accurately than just sampling the non-missing values of `weight`.
- ▶ We will assume from here on that the data consists entirely of continuous variables. Allowing other types of variables is more complex, but follows similar methods.



Patterns of Missing Data

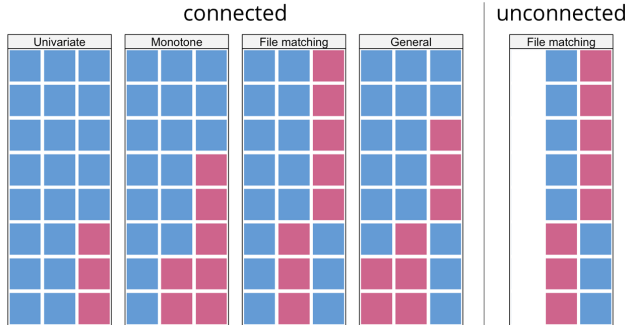
Let the data be represented by the $n \times p$ matrix $\mathbf{X}$. In the presence of missing data $\mathbf{X}$ is partially observed. Notation X_j is the j^{th} column in $\mathbf{X}$, and $\mathbf{X}_{-j}$ are all columns except X_j .

The **missing data pattern** of $\mathbf{X}$ is the $n \times p$ binary response matrix R indicating the “positions” of missing data in $\mathbf{X}$.

The actually observed data consist of X_{obs} and R .



Types of missing data patterns



- ▶ A file matching pattern is called connected when it is possible to travel to all blue cells by horizontal or vertical moves (rearrangement of columns and rows allowed.)
- ▶ Connected patterns are needed to identify unknown parameters. For example, in order to be able to estimate a correlation coefficient between two variables.



General strategies for imputing multivariate data

1. **Monotone data imputation:** For monotone missing data patterns, imputations are created by a sequence of univariate imputations based on multivariate methods (see below)
2. **Joint modeling:** For general patterns, imputations are drawn from a *explicitly* specified p -multivariate model fitted to the data.
3. **Fully conditional specification**, also known as *chained equations* and *sequential regressions*: For general patterns, a p -multivariate model is *implicitly* specified by a set of p conditional univariate models. Imputations are created by drawing from iterated conditional models.



Monotone Data Imputation with Multivariate Regression

- ▶ We can take the elements with complete data and fit a regression model to predict the missing values for another variable.
- ▶ The type of regression can be any regression method but is often multiple linear regression.
- ▶ E.g. Assume we have complete data ($n=10000$) for age, race, gender and BMI. The variable `Pulse` has 1437 missing values.
- ▶ We can fit a linear regression model

```
Pulse ~ BMI + age + race + gender
```

using the 8563 elements with complete data.

- ▶ The predicted missing values should then be **simulated with an error term**:
If $\hat{y}_i$ is the predicted value from the regression model, then the imputed value should be $\tilde{y}_i = \hat{y}_i + \epsilon_i$, where ϵ_i is a random normal simulation with the model residual variance.



Typical Problems imputing Multivariate Data

- ▶ For a given variable X_j , the other variables $\mathbf{X}_{-j}$ used in the imputation model may themselves be incomplete.
- ▶ Circular dependencies: X_1 depends on X_2 and X_2 depends on X_1 .
- ▶ Variables can be of different types (e.g., binary, unordered, ordered, continuous), thereby making the application of theoretically convenient models, such as the multivariate normal, theoretically inappropriate.
- ▶ The relation between X_j and $\mathbf{X}_{-j}$ could be complex, e.g., nonlinear, or subject to censoring processes.
- ▶ Imputation can create impossible combinations (e.g., pregnant fathers), or destroy deterministic relations in the data (e.g., sum scores, BMI).
- ▶ Imputations can be nonsensical (e.g., body temperature of the dead).

source: van Buuren et al., 'Mice': Multivariate Imputation by Chained Equations in R. Journal of Statistical Software 45 (3): 1-67. (2011)



Imputation with Multivariate Regression (cont)

Remarks:

- ▶ As with *Mean/Variance Simulation*, simulating the error term avoids a reduction in the variance of that variable. Remember, our aim is not to get the best prediction for the missing value but to preserve the overall properties of the complete data set.
- ▶ The regression method is not restricted to continuous variables.
- ▶ With **categorical variables** predicted values from (multinomial) logistic regression or tree models for classification.



When more than one variable has missing values

The *multivariate regression method* easily adapts itself to imputing multiple variables, by looping over the variables with missing values.

Suppose $X_1, \dots, X_k$ are variables with some missing values and $X_{k+1}, \dots, X_p$ are complete.

- ▶ Fit the regression $X_1 \sim X_{k+1} + X_{k+2} + \dots + X_p$, and impute all the missing values in X_1 including a simulated error term.
- ▶ Then fit the regression $X_2 \sim X_1 + X_{k+1} + X_{k+2} + \dots + X_p$, and impute all the missing values in X_2 with simulated error term.
- ▶ Continue until all variables $X_1, \dots, X_k$ have been completed.

Problem: if X_1 is correlated with X_2 then this is not considered when imputing X_1 , but is considered when imputing X_2



Solution

So in practice the following approach is used:

- ▶ First, use a simple univariate method such as *direct random sampling* to complete each variable. Keep a record of the “positions” of the original missing values.
- ▶ Fit the regression $X_1 \sim X_2 + X_3 + \dots + X_p$, using all elements for which X_1 was *not* imputed.
- ▶ Re-impute all the originally missing values in X_1 including a simulated error term.
- ▶ Then fit the regression $X_2 \sim X_1 + X_3 + X_4 + \dots + X_p$, using all elements for which X_2 was *not* imputed and re-impute.
- ▶ Repeat for $X_3, \dots, X_k$.

Because the X_1 imputations were based on some *direct random sampled* values of $X_2, \dots, X_k$, it is best to repeat this loop a couple more times.



MICE: Multiple Imputation by Chained Equations

Concept:

- ▶ Imputation takes place on a variable-by-variable basis.
- ▶ A specification of an imputation model for each incomplete variable is required.
- ▶ Imputations are created per variable in an iterative fashion.
- ▶ The observed p -multivariate data distribution is specified through a set of p conditional distributions for the individual variables.
- ▶ Gibbs sampling (GS) is applied which is a Bayesian simulation technique that samples from the conditional distributions in order to obtain samples from the joint distribution².

²more details later in this course



The MICE algorithm

1. Initialisation: for each X_j , fill in starting imputations $\dot{X}_j^0$ by random draws from X_j^{obs} .
2. Specify an conditional imputation model for each variable X_j

$$P(X_j^{mis} | X_j^{obs}, \mathbf{X}_{-j}, R, \theta_j)$$

where θ_j are the parameters specific for the j -th conditional density.

3. M iterations: Repeat for $t = 1, \dots, M$
 - a) Repeat for all variables X_j with missing values:
 - i) Define $\dot{\mathbf{X}}_{-j}^t = (\dot{X}_1^t, \dots, \dot{X}_{j-1}^t, \dot{X}_{j+1}^t, \dots, \dot{X}_p^t)$ as the currently complete data except X_j .
 - ii) Estimate $P(\theta_j^t | X_j^{obs}, \dot{\mathbf{X}}_{-j}^t, R)$ based on the specific imputation model.
 - iii) Draw $\dot{\theta}_j^t \sim P(\theta_j^t | X_j^t, \dot{\mathbf{X}}_{-j}^t, R)$.
 - iv) Draw imputations $\dot{X}_j^{t+1} \sim P(X_j^{mis} | X_j^{obs}, \dot{\mathbf{X}}_{-j}^t, R, \dot{\theta}_j^t)$
 - b) End repeat j .
4. End repeat t .



MICE in R

R package `mice`: Multivariate Imputation by Chained Equations, uses Gibbs sampling.

- ▶ This package easily allows for multiple realisations of the complete data set.
- ▶ The data analysis (if using R) can then be repeated on each of the realisations (using one command) and an aggregate model obtained.

Example:

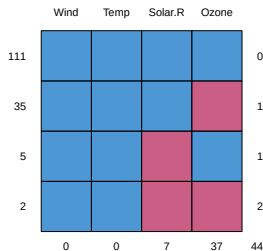
```
library(mice)
data("airquality")
airquality <- airquality[, c("Ozone", "Solar.R", "Wind", "Temp")]
md.pattern(airquality)      # missing data pattern
```

	Wind	Temp	Solar.R	Ozone	
111	1	1	1	1	0
35	1	1	1	0	1
5	1	1	0	1	1
2	1	1	0	0	2
	0	0	7	37	44



Example `airquality`: pattern of missing data

Missing data pattern as graph:

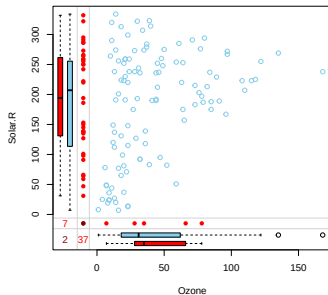


- ▶ no missings for `Wind`, `Temp`, `Month` and `Day`
- ▶ `Solar.R` has 7 missing values.
- ▶ `Ozone` has 37 missing values.
- ▶ Two rows have both `Solar.R` and `Ozone` missing.



Example airquality: pattern of missing data

```
library(VIM)
marginplot(airquality[, c("Ozone", "Solar.R")])
```



If the assumption of MCAR data is correct, the red and blue box plots are expected to be very similar.



Example airquality: MICE

```
airqualityImp <- mice(airquality, m = 5, maxit = 50, method = "norm")
```

Arguments:

- ▶ number of imputations/realisations: `m`
- ▶ maximum iterations: `maxit`
- ▶ *Bayesian linear regression*: `method = "norm"`

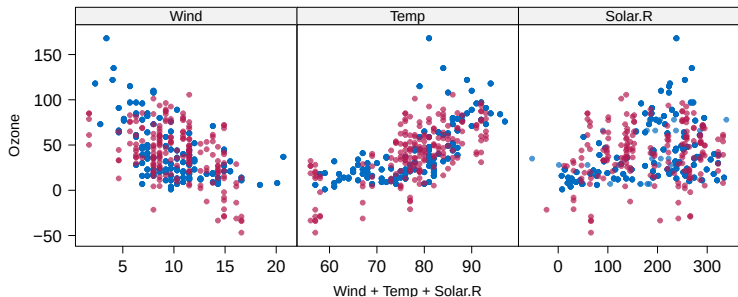
These are the imputed values for `ozone`. The 5 columns are the replications, and the row numbers are given.

```
head(airqualityImp$imp$ozone) |> round(2)
```

	1	2	3	4	5
5	18.74	26.64	27.09	-33.47	32.39
10	4.62	26.55	40.26	14.68	37.87
25	-46.82	13.40	-34.13	-33.31	-29.03
26	15.51	-28.35	-29.21	13.17	-2.11
27	15.58	-21.49	19.39	20.85	28.17
32	52.33	11.38	59.31	61.24	35.29



Example airquality: inspecting the imputed values



What we would like to see is that the shape of the red points (imputed) matches the shape of the blue ones (observed). The matching shape tells us that the imputed values are indeed *plausible* values.



Example airquality: why again?

Q: Why did we impute the missing data again?



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A: To perform an analysis without losing observations.



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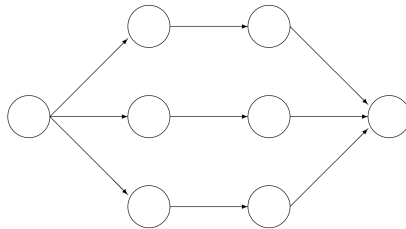
Example:

- ▶ plan was to fit a multiple linear regression model to `Temperature` using the variables `Ozone`, `Solar.R` and `Wind`.
- ▶ Now that we have **complete data** we can fit this regression model, but we have **5 realisations**.
- ▶ Rather than choose one of the 5, we can fit the **model to each of the realisations** and “pool” (average) the results. This should be done using the function `mice::pool()`.



Pooling

Scheme of main steps in multiple imputation.



Incomplete data Imputed data Analysis results Pooled result

taken from: <https://stefvanbuuren.name/fimd/sec-nutshell.html>



Example airquality: why again?

analysis **with** imputed values:

```
lm_imp <- with(airqualityImp, lm(Temp ~ Ozone + Solar.R + Wind))
summary(pool(lm_imp))
```

	term	estimate	std.error	statistic	df	p.value
1	(Intercept)	71.4738	2.70678	26.41	136.5	1.70e-55
2	Ozone	0.1736	0.02304	7.54	123.8	8.90e-12
3	Solar.R	0.0112	0.00737	1.51	49.2	1.36e-01
4	Wind	-0.2934	0.19703	-1.49	141.8	1.39e-01

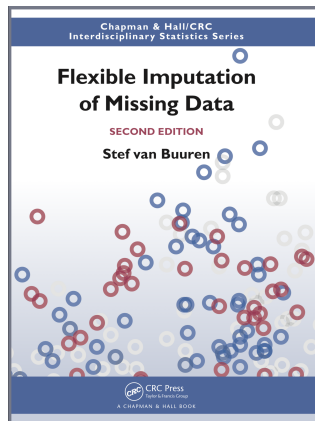
analysis **without** imputed values:

```
lm_naOmit <- with(na.omit(airquality), lm(Temp ~ Ozone + Solar.R + Wind))
printCoefmat(summary(lm_naOmit)$coefficients, signif.stars = FALSE)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	72.41858	3.21552	22.52	< 2e-16
Ozone	0.17197	0.02639	6.52	0.0000000024
Solar.R	0.00728	0.00768	0.95	0.35
Wind	-0.32294	0.23326	-1.38	0.17



Further Reading



- ▶ <https://stefvanbuuren.name/fimd/>
- ▶ van Buuren et al., “Mice”: *Multivariate Imputation by Chained Equations in R*. Journal of Statistical Software 45 (3): 1–67. (2011) (<https://www.jstatsoft.org/article/view/v045i03>)